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AutoGen Practical Quiz

🔹 Section A: Scenario-Based MCQs (Choose the BEST answer)

1. Multi-Agent Debugging System

You are building an AutoGen system where:

- Agent A: Writes Python code
- Agent B: Reviews and debugs
- Agent C: Executes code

The system is failing because Agent B is not correcting errors properly.

What is the BEST improvement?

A. Remove Agent B
B. Add prompt constraints to Agent B for structured feedback
C. Merge all agents into one
D. Increase token limit

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2. Conversational Agent Memory

A chatbot built using AutoGen forgets previous user inputs in a multi-turn conversation.

What should you implement?

A. Stateless API calls
B. Conversation memory or context window management
C. Reduce number of agents
D. Use only system prompts

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3. Agent Role Design

You created two agents:

- ResearchAgent
- SummaryAgent

Both are generating similar outputs.

What is the issue?

A. Token limit exceeded
B. Poor role separation in system prompts
C. API failure
D. Execution error

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4. Infinite Loop Problem

Two agents keep responding to each other indefinitely.

What is the BEST fix?

A. Increase temperature
B. Add termination condition or max_turns
C. Add more agents
D. Reduce prompt size

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5. Tool Usage in AutoGen

An agent needs to fetch real-time stock data.

What is the correct approach?

A. Hardcode stock values
B. Integrate tool/API calling capability
C. Use static prompts
D. Disable conversation

◆ Section B: Code-Based Scenarios

6. Identify the Issue

```
assistant = AssistantAgent(name="assistant")
user = UserProxyAgent(name="user",code_execution_config={"work_dir": "coding",
"use_docker": False}
)

user.initiate_chat(assistant, message="Write a function to sort a list")
```

The assistant gives code but it is never executed.

What is missing?
code_execution_config={"work_dir": "coding", "use_docker": False}

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7. Fix the Collaboration Flow

```
agent1 = AssistantAgent(name="coder")
agent2 = AssistantAgent(name="reviewer")

agent1.initiate_chat(agent2, message="Write code for factorial")
```

Problem: Reviewer is not giving meaningful feedback.

What change would you make in agent design?
Weak prompting the is major reason for meaningful feedback.
provide structural prompting.

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8. Execution Agent Issue

```
user_proxy = UserProxyAgent(
    name="user",
    code_execution_config=True
)
```

Why will this fail in a coding workflow?
This fails because code_execution_config=False disables execution

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9. Add a Third Agent

You have:

- Coder Agent
- Executor Agent

Now you want validation before execution.

What type of agent should you introduce and where in flow?
Coder → Reviewer → Executor. This ensures correctness before running code.
So add a new agent.

🔹 Section C: Design-Based Questions (Short Answers)

10. Multi-Agent Architecture Design

Design an AutoGen system for:
👉 "Automated Resume Screening"

Include:

- Agent roles
- Interaction flow

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11. Conversational Agent Improvement

A chatbot gives generic answers and lacks personalization.

How would you improve it using AutoGen?
Improve personalization by adding memory + user profiling agent. This helps track user preferences and provide context-aware responses.

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12. Agent Collaboration Strategy

You are building a **Research Assistant** using:

- Web search
- Summarization
- Report generation

How will agents collaborate step-by-step?
1. a Search Agent gathers data

2. a Summarizer Agent condenses it.
3. a Report Generator creates structured output.

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13. Error Handling Scenario

During execution, one agent generates invalid Python code.

How should the system handle this collaboratively?
If invalid code is generated, the Executor sends error feedback to the Coder. The system loops until the code is corrected.

🔹 Section D: Advanced Practical Scenarios

14. Real-World Workflow

Build a system for:
👉 “Automated Data Analysis from CSV”

What agents will you create and what will each do?
1. Agents include Data Loader (reads CSV),
2. Analyst (performs analysis)
3. Visualizer (creates charts).
The flow is load → analyze → present insights.

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15. Prompt Engineering Challenge

An agent gives vague answers.

Rewrite the agent’s system prompt to:

- Improve clarity
- Ensure structured output
  Prompt:Provide step-by-step answer with bullet points and examples.

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16. Multi-Agent Decision System

You want agents to **vote on best solution**.

How will you design this?
Use multiple agents to generate solutions, then add a **Voting Agent**. It selects the best output based on majority or scoring.

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17. Scaling Issue

Your AutoGen system becomes slow with multiple agents.

What optimizations would you apply?
Optimize by reducing unnecessary agents, limiting message history, caching responses, and using efficient models. This improves speed and cost.

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18. Security Scenario

An agent executes user-provided code.

What safeguards will you implement?
Add safeguards like Docker, input validation, restricted permissions

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🔹 Section E: Mini Case Study

19. End-to-End System Design

Build an AutoGen system for:

👉 **"AI Tutor for B.Tech Students"**

Requirements:

- Answer doubts
- Generate quizzes
- Evaluate answers

Design:

- Agents involved
- Collaboration flow
- Memory handling
  1. Agents: Tutor Agent (explains concepts)

2. Quiz Generator (creates questions)
3. Evaluator (checks answers)
4. Memory Agent (tracks student progress).
question → explanation → quiz → evaluation → feedback loop.

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20. Debug This Workflow

Flow:

1. User → Planner Agent
2. Planner → Coder
3. Coder → Executor
4. Executor → User

Problem: Output is incorrect and no validation happens.
The problem is missing validation.
Add a Reviewer Agent between Coder and Executor.
Planner → Coder → Reviewer → Executor → User